Estimating Square & Cube Roots

Guided Notes
NC.8.NS.2 · Unit 1

Name: ____________________________ Date: ________________ Period: ______

PART 1 — Place a Root on a Number Line

A non-perfect square root or cube root always falls between two consecutive whole numbers. Fill in the steps:

Step 1. Find the two consecutive perfect ________ (squares / cubes) that your number falls between.

Step 2. Take the ________ root of each — those are your two consecutive ________ (whole numbers).

Step 3. Decide: is your number closer to the ________ perfect value or the ________ one?

Step 4. Place it on the number line ________.

Try it: Estimate $\sqrt{45}$.

____ < 45 < ____ so $\sqrt{45}$ is between ____ and ____ , and closer to ____ .

↑ Label the two whole numbers on the ends, then mark where $\sqrt{45}$ belongs.

PART 2 — The 4-Step Algorithm (to the nearest tenth)

$\sqrt{n} \approx$ ________ + (n − ________) ÷ (________ − ________)

(lower root) (lower²) (upper²) (lower²) · for a cube root, use cubes: lower³ and upper³.

Square root — estimate $\sqrt{45}$:

Step 1	____ < 45 < ____ → √____ = ____ , √____ = ____
Step 2	(45 − ____) ÷ (____ − ____) = ____ ÷ ____
Step 3	Divide: ____ ÷ ____ ≈ ________
Step 4	____ + ________ = ________ → $\sqrt{45} \approx$ ________ (Calculator: 6.708…)

Cube root — estimate $\sqrt[3]{30}$: (same steps, but cube the whole numbers)

Step 1	____ < 30 < ____ → ∛____ = ____ , ∛____ = ____
Step 2	(30 − ____) ÷ (____ − ____) = ____ ÷ ____
Step 3	Divide: ____ ÷ ____ ≈ ________
Step 4	____ + ________ = ________ → $\sqrt[3]{30} \approx$ ________ (Calculator: 3.107…)

Practice — estimate each to the nearest tenth:

$\sqrt{50} \approx$ ________	$\sqrt{60} \approx$ ________	$\sqrt[3]{100} \approx$ ________	$\sqrt[3]{40} \approx$ ________

PART 3 — Ordering a Mixed Set (convert everything to a decimal)

To compare rationals, irrationals, fractions and mixed numbers, turn everything into a decimal first, then line them up.

Order from least to greatest: √5 , 2 1/4 , 7/3 , 2.1

Number	How to convert	Decimal
√5	4-step algorithm	________
2 1/4	2 + (1 ÷ 4)	________
7/3	7 ÷ 3	________
2.1	already a decimal	________

Least → Greatest: ________ < ________ < ________ < ________

Reminder. Mixed number → decimal: whole + (numerator ÷ denominator). Example: 3 2/5 = 3 + (2 ÷ 5) = ________.

⚠ Error Analysis

To order 1.5 , √2 , 7/4 , √3 , a student wrote 1.5 < √2 < √3 < 7/4 . Convert first, then fix it:

1.5 = ______ , √2 ≈ ______ , 7/4 = ______ , √3 ≈ ______

What did the student miss? ________________________

Correct order: ________ < ________ < ________ < ________